ds homework

1)

```
#include<stdio.h>
int main(){
    int i,j,m,n;
    scanf("%d %d",&m,&n);
    int a[m][n],b[m][n],c[m][n];
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        scanf("%d",&a[i][j]);
    }
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        scanf("%d",&b[i][j]);
    }
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        c[i][j]=a[i][j]+b[i][j];
    }
    printf("Added matrices:\n");
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        printf("%d ",c[i][j]);
        printf("\n");
    }
}
```

Output:

```
2 3
1 2 3
4 5 6
7 8 9
10 11 12
Added matrices:
8 10 12
14 16 18
```

2)

```
#include<stdio.h>
int main(){
    int i,n;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
    scanf("%d",&a[i]);
    for(i=0;i<n;i++)
    printf("%d ",a[i]);
    printf("\n");
    printf("Reversed array:\n");
    for(i=n-1;i>=0;i--)
```

```
    printf("%d ",a[i]);
}
```
Output:
5
10 20 30 40 50
10 20 30 40 50
Reversed array:
50 40 30 20 10

3)
```
#include <stdio.h>
int main() {
    int i, j, n, count = 0;

    scanf("%d", &n);
    int a[n];

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n; i++) {
        for(j = i + 1; j < n; j++) {
            if(a[i] == a[j]) {
                count++;
                break;
            }
        }
    }
    printf("Count of Duplicates:\n");
    printf("%d", count);

    return 0;
}
```
Output:
5
1 2 2 3 3
Count of Duplicates:
2

4)
```
#include <stdio.h>

int main() {
    int i, j, n, count;

    scanf("%d", &n);
    int a[n];

    for(i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);

    for(i = 0; i < n; i++)
    {
        count=0;
        for(j = 0 ; j < n; j++) {
            if(a[i] == a[j]) {
                count++;

            }
        }
        if(count==1)
        printf("%d ",a[i]);
    }


    return 0;
}
Output:
5
1
2
2
3
5
1 3 5

5)
#include<stdio.h>
int main(){
    int n,i,j,ele,pos;
    scanf("%d",&n);
    int a[n];
    printf("enter sorted array:");
    for(i=0;i<n;i++)
    scanf("%d",&a[i]);
    scanf("%d",&ele);
    for(i=0;i<n;i++){
        if(ele<a[i]){
        pos =i;
        break;}
    }
    if(i==n)
    pos =n;
    for(i=n-1;i>=pos;i--){
        a[i+1]=a[i];
    }
    a[pos]=ele;
    n++;
    for(i=0;i<n;i++)
```

```
    printf("%d ",a[i]);
}
```

Output:
5
enter sorted array:10
20
30
40
50
25
10 20 25 30 40 50

6)

```
#include <stdio.h>

int main() {
    int n, i, ele, pos;

    scanf("%d", &n);

    int a[n + 1];


    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);


    scanf("%d", &ele);


    scanf("%d", &pos);

    pos = pos-1;

    for(i = n - 1; i >= pos; i--) {
        a[i + 1] = a[i];
    }

    a[pos] = ele;
    n++;


    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output:
5
12

34
45
67
89
56
3
12 34 56 45 67 89

7)
```
#include<stdio.h>
int main(){
    int i,j,k,r1,c1,r2,c2;
    scanf("%d %d",&r1,&c1);
    scanf("%d %d",&r2,&c2);
    int A[r1][c1],B[r2][c2],C[r1][c2];
    for(i=0;i<r1;i++){
        for(j=0;j<c1;j++)
        scanf("%d",&A[i][j]);
    }

    for(i=0;i<r2;i++){
        for(j=0;j<c2;j++)
        scanf("%d",&B[i][j]);
    }
    for(i=0;i<r1;i++){
        for(j=0;j<c2;j++){
            C[i][j]=0;

        for(k=0;k<c1;k++)
            C[i][j]+=A[i][k]*B[k][j];
        }
    }
    for(i=0;i<r1;i++){
        for(j=0;j<c2;j++)
        printf("%d ",C[i][j]);
        printf("\n");
    }
}
```
Output:
2 2
2 2
1
2
3
4
5
6
7
8
19 22

43 50

8)
```
#include<stdio.h>
int main(){
    int i,j,m,n;
    scanf("%d",&m);
    scanf("%d",&n);
    int a[m][n],b[m][n];
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        scanf("%d",&a[i][j]);
    }
    printf("Input Matrix\n");
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        printf("%d ",a[i][j]);
        printf("\n");
    }
    printf("Transpose matrix:\n");
    for(j=0;j<n;j++){
        for(i=0;i<m;i++)
        printf("%d ",a[i][j]);
        printf("\n");
    }
}
```
Output:
```
2
2
1
2
3
4
Input Matrix
1 2
3 4
Transpose matrix:
1 3
2 4
```

9)
```
#include<stdio.h>
int main(){
    int i,j,m,n,sum=0;
    scanf("%d",&m);
    scanf("%d",&n);
    int a[m][n],b[m][n];
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        scanf("%d",&a[i][j]);
```

```
    }
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        printf("%d ",a[i][j]);
        printf("\n");
    }
    for(i=0;i<m;i++){
        for(j=0;j<n;j++)
        if(i==j)
        sum+=a[i][j];
    }
    printf("Diagonal Sum:%d",sum);
}
```

Output:

```
3
3
1 2 3
4 5 6
7 8 9
1 2 3
4 5 6
7 8 9
Diagonal Sum:15
```

10)

```
#include <stdio.h>

int main() {
    int i, j, m, n, identity = 1;

    scanf("%d", &m);
    scanf("%d", &n);

    int a[m][n];

    for(i = 0; i < m; i++) {
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    }

    for(i = 0; i < m; i++) {
        for(j = 0; j < n; j++)
            printf("%d ", a[i][j]);

        printf("\n");
    }

    if(m != n) {
        identity = 0;
    }
```

```
    else {
        for(i = 0; i < m; i++) {
            for(j = 0; j < n; j++) {

                if(i == j) {
                    if(a[i][j] != 1)
                        identity = 0;
                }
                else {
                    if(a[i][j] != 0)
                        identity = 0;
                }
            }
        }
    }

    if(identity == 1)
        printf("It is an identity matrix");
    else
        printf("It is not an identity matrix");

    return 0;
}
```

Output:

```
3
3
1 0 0
0 1 0
0 0 1
1 0 0
0 1 0
0 0 1
It is an identity matrix

3
3
1 0 2
3 4 0
7 8 1
1 0 2
3 4 0
7 8 1
It is not an identity matrix
```

11)

```
#include <stdio.h>

int main() {
    int i, v = 0, c = 0, d = 0, w = 0;
    char str[100];
```

```
    fgets(str, 100, stdin);

    for(i = 0; str[i] != '\0'; i++) {
        if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||
           str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
            v++;
        else if((str[i]>='a'&&str[i]<='z')||(str[i]>='A'&&str[i]<='Z'))
            c++;
        else if(str[i]>='0'&&str[i]<='9')
            d++;
        else if(str[i]==' '||str[i]=='\t'||str[i]=='\n')
            w++;
    }

    printf("Vowels: %d\n", v);
    printf("Consonants: %d\n", c);
    printf("Digits: %d\n", d);
    printf("White spaces: %d", w);

    return 0;
}
Output:
hello world 123
Vowels: 3
Consonants: 7
Digits: 3
White spaces: 3

12)
#include <stdio.h>

int main() {
    int i,j,count=0;
    char str[100],ch;

    fgets(str, 100, stdin);
    scanf("%c",&ch);
    for(i=0;str[i]!='\0';i++){
        if(str[i]==ch)
        count++;
        }

    printf("%d",count);
}
Output:
banana
n
2
```

13)
```
#include <stdio.h>

int main() {
    int i,j,count=0;
    char str[100];

    fgets(str, 100, stdin);
    for(i=0;str[i]!='\0';i++){
        if(str[i]!=' '&&(i==0||str[i-1]==' '))
        count++;
    }
    printf("%d",count);
}
```

Output:
```
hello  i am sebastina
4
```

14)
```
#include<stdio.h>
#include<string.h>
void ins_sort(int n, char arr[n][50]);
int main(){
    int i ,n;
    scanf("%d",&n);
    char str[n][50];
    for(i=0;i<n;i++){
        scanf("%s",str[i]);

    }
    ins_sort(n,str);
    printf("sorted array:\n");
    for(i=0;i<n;i++)
    printf("%s\n",str[i]);

}
void ins_sort(int n , char arr[n][50]){
    int i,j;
    char temp[50];
    for(i=1;i<n;i++){
    strcpy(temp,arr[i]);
    j=i-1;
    while(j>=0 && strcmp(arr[j],temp)>0){
        strcpy(arr[j+1],arr[j]);
        j=j-1;
    }
    strcpy(arr[j+1],temp);
    }
}
```

```
Output:
5
mango
apple
banana
kiwi
guava
sorted array:
apple
banana
guava
kiwi
mango
```

15)

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

    fgets(str, 100, stdin);

    while(str[i] != '\0') {
        if(str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
        else if(str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;

        i++;
    }

    printf("%s", str);

    return 0;
}
```

```
Output:
Hello AmEriCa
hELLO aMeRIcA
```